

SHA TIN, Hong Kong

WMO: 450390

Lat: 22.4025N

Lon: 114.2100E

Elev: 13

StdP: 101.17

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	9.1	-4.2	2.7	13.1	-1.1	3.4	13.0	5.4	13.6	5.0	14.3	2.4	340	0.304

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	33.9	26.5	33.0	26.5	32.3	26.4	27.7	31.5	27.5	31.1	27.1	30.7	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.4	30.0	26.5	22.0	29.8	26.2	21.6	29.5	89.2	31.8	87.4	31.1	86.0	30.8	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.3	4.7	5.8	36.1	2.4	0.7	4.1	36.6	2.7	37.0	1.4	37.4	-0.3	38.0		
			2.3	29.7	2.1	1.5	0.7	30.7	-0.5	31.6	-1.7	32.4	-3.3	33.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	15.9	17.1	19.3	23.0	26.3	28.5	29.3	29.1	28.4	25.8	22.0	17.6
	DBStd	5.46	3.17	3.83	3.23	2.78	2.23	1.60	1.36	1.42	1.47	1.91	2.83	3.07
	HDD10.0	4	2	2	0	0	0	0	0	0	0	0	0	1
	HDD18.3	236	87	62	28	3	0	0	0	0	0	0	6	51
	CDD10.0	4947	184	199	287	389	507	554	597	591	551	489	362	236
	CDD18.3	2135	11	26	57	142	248	304	339	332	301	231	117	27
	CDH23.3	21012	14	55	149	744	2265	3591	4300	4106	3448	1841	460	39
	CDH26.7	7542	0	1	9	124	659	1395	1899	1732	1274	405	42	1
Wind	WSAvg	2.4	2.2	2.3	2.4	2.4	2.4	2.7	2.7	2.3	2.2	2.3	2.3	2.3
Precipitation	PrecAvg	1924	29	37	54	129	256	406	322	321	236	69	36	29
	PrecMax	2876	246	171	208	586	563	1344	1006	836	532	417	254	191
	PrecMin	1106	0	0	4	0	41	97	75	31	52	0	0	0
	PrecStd	479	46	47	51	119	119	261	204	230	140	102	53	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	26.5	27.7	30.4	33.2	33.3	35.5	35.3	34.6	32.2	29.3	26.1
		MCWB	19.2	21.6	22.6	24.6	26.3	26.4	27.2	26.7	26.1	24.8	22.8	20.0
	2%	DB	23.0	25.1	26.4	29.1	31.8	32.6	33.6	33.9	33.0	30.9	27.8	24.4
		MCWB	18.3	21.2	22.1	24.2	26.1	26.5	26.8	26.6	26.1	24.2	22.1	18.9
	5%	DB	21.2	23.7	25.1	28.1	30.8	32.0	32.8	32.9	32.1	29.7	26.7	23.0
		MCWB	17.1	20.0	21.6	23.8	25.9	26.6	26.6	26.6	25.6	23.4	21.6	18.1
	10%	DB	20.0	22.1	23.6	27.1	29.7	31.2	32.0	32.0	31.2	28.8	25.8	21.7
		MCWB	16.5	19.1	20.3	23.5	25.6	26.5	26.5	26.4	25.5	23.2	21.4	17.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	23.0	23.6	25.7	27.2	27.9	28.6	28.5	27.7	26.8	24.4	21.6
		MCDB	23.4	25.6	26.3	28.8	30.7	31.1	31.7	32.8	31.5	29.9	27.6	24.0
	2%	WB	19.2	21.9	22.9	25.2	26.9	27.6	27.8	27.7	27.1	25.8	23.7	20.4
		MCDB	21.8	24.2	25.8	27.9	30.1	30.7	31.7	31.7	30.9	28.8	26.5	22.8
	5%	WB	18.2	20.7	22.1	24.6	26.5	27.2	27.3	27.3	26.7	25.1	23.1	19.5
		MCDB	20.4	23.0	24.4	27.3	29.8	30.4	31.0	31.1	30.3	28.4	25.7	22.2
	10%	WB	17.2	19.5	21.1	24.0	26.1	27.0	27.0	27.0	26.3	24.5	22.4	18.2
		MCDB	19.4	21.4	23.0	26.6	29.3	30.1	30.6	30.6	29.7	27.8	25.0	21.1
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.2	5.1	4.9	4.6	4.4	5.0	5.3	5.3	5.3	5.3	5.9
		MCDBR	7.1	7.4	7.3	6.6	6.0	5.4	6.2	6.9	6.5	6.3	6.2	7.2
		MCWBR	3.4	3.7	3.7	2.9	2.1	1.8	2.1	2.4	2.2	2.6	2.7	3.3
	5% WB	MCDBR	5.9	6.0	6.2	5.6	5.1	4.5	5.5	5.9	5.8	5.0	4.9	5.9
	MCWBR	3.5	3.5	3.5	3.0	2.2	1.8	2.2	2.5	2.2	2.4	2.5	3.4	
Clear-Sky Solar Irradiance	taub	0.582	0.635	0.763	0.740	0.578	0.547	0.535	0.600	0.633	0.676	0.575	0.568	
	taud	1.800	1.727	1.519	1.601	1.988	2.087	2.116	1.917	1.820	1.680	1.882	1.855	
	Ebn at Noon	678	674	612	640	749	766	775	728	694	633	680	671	
	Edh at Noon	198	226	290	271	183	164	159	195	211	234	181	182	
All-Sky Solar Radiation	RadAvg	3.07	3.12	3.36	3.74	4.47	4.59	5.34	5.01	4.72	4.39	3.56	3.19	
	RadStd	0.50	0.49	0.45	0.57	0.61	0.69	0.56	0.46	0.35	0.44	0.40	0.47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	+0.25	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page